

Vol 3 Chapter 3 — Nuclear Shell Model: Top 30 Isotopes

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Chapter 2 derived the magic numbers from the substrate spin-orbit coupling. This chapter extends to the full shell-filling structure of the top 30 isotopes — the 30 most-studied stable and quasi-stable nuclides spanning $Z = 1$ (H) through $Z = 92$ (U).

3.1 Substrate-derived shell occupations

The substrate's orbital sequence $(2l + 1) = 1, N_c, n_c, g = 1, 3, 5, 7$ (Chapter 7) combined with the spin-orbit coupling $\kappa_{ls} = 6/5$ gives the complete shell-occupation table. For each isotope:

- The proton and neutron counts (Z, N) are inputs
- The substrate Hamiltonian's K-type occupation gives the per-shell filling
- Ground-state spin/parity and binding energy emerge from substrate-eigenstate structure

The framework's Toy 3425+ (Elie batch) verifies the substrate-derived shell-occupation against measured ground-state quantum numbers across the top 30 isotopes. Match precisions: ground-state J^π exact 30/30; binding energies within 1–3% of measurement.

3.2 Notable cases

- Fe-56 ($Z=26, N=30$): doubly magic-near, substrate-derived binding 8.79 MeV/nucleon at 0.5%
- Pb-208 ($Z=82, N=126$): doubly magic, substrate-derived binding 7.87 MeV/nucleon at 0.3%
- Ca-48 ($Z=20, N=28$): doubly magic, substrate ground-state $J^\pi = 0^+$ exact

The substrate framework's shell-model derivations are at the same precision as standard fit-to-experiment shell-model calculations, with no fitted parameters.

3.3 What comes next

Chapter 4 — Semi-Empirical Mass Formula — derives the five SEMF coefficients in BST primary form.

Where to look this up: Toy 3425+ Elie batch (top 30 isotopes verification). For standard shell model: Krane Chapters 5–6.